Vertex Formula

1 The x-coordinate of the vertex of the parabola occurs at $x = \frac{-b}{2a}$. The y-coordinate of the vertex is found by substituting $x = \frac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x.

2 Axis of symmetry

3

a $x = 0, 6$ **b** $(3, 9)$

4

a Concave up **b** $x = 9$ **c** $y = -71$

5

a Concave up **b** $x = 1, 5$

c $y = 5$ **d** $x = 3$

e $(3, -4)$

6

a Concave down **b** $x = -4, 2$

c $y = 8$ **d** $x = -1$

e $(-1, 9)$

7

a Concave down **b** $x = -4, -2$

c $y = -8$ **d** $x = -3$

e $(-3, 1)$

8

a $x = -2$ **b** -1

c $(-2, -1)$ **d** $(-2, -2)$

e

i Yes **ii** No **iii** Yes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

9

a $x = 14, 6$ **b** $x = 10$ **c** Concave down **d** $y = 16$

10 $(1, -12)$

11 $y = 0$

12

a -0.087 **b** -1.24

13 $x = \frac{1}{4}$

14

a Vertex of $P_1 = (-2, 2)$
Vertex of $P_2 = (2, 2)$

b 4

15

a $x = -3$ **b** -5

Applications

16

a $s = 12$ **b** 337

17

a $t = 3$ **b** 122 units

© ClipSAT: By Mr. Mohamed Abdallah +966597688647